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Bootstrap, CLT, and Normal Approximation

1) $$SE(\bar{X}) = \frac{S}{\sqrt{n}} = \frac{10}{\sqrt{100}} = 1$$

$$z = \frac{\bar{X} - \mu_0}{SE} = \frac{52 - 50}{1} = 2$$

The two sided p value is $2P(Z > 2)$

$$= 2\left(1 - \Phi(2.00)\right) = 2(0.02275)$$

$$= 0.0455$$

for $\alpha = 0.10$, critical value $= 1.645$

which is $< |z| = 2$

$\therefore$ reject H_0

for $\alpha = 0.05$, critical value $= 1.96$

which is $< |z| = 2$

$\therefore$ reject H_0

For $\alpha = 0.01$, critical value $= 2.576$

which is $> |z| = 2.00$

$\therefore$ fail to reject H_0

95% CI

$$\bar{x} \pm 1.96 \cdot SE = 52 \pm 1.96(1)$$

$$= (50.04, 53.96)$$

2) $$z = \frac{\hat{S} - 0}{\hat{SE}(\hat{S})} = \frac{3.1}{1.3} \approx 2.38$$

$$2P(Z > 2.38) \approx 2(0.0086) \approx 0.017$$

level	Margin of error	interval
90%	$1.645 \times 1.3 = 2.14$	$(0.96, 5.24)$
95%	$1.96 \times 1.3 = 2.55$	$(0.55, 5.65)$
99%	$2.576 \times 1.3 = 3.35$	$(-0.25, 6.45)$

Critical value at $\alpha = 0.10 < |z|$ and 0 is not in the CI $\therefore$ Reject H_0

Critical value at $\alpha = 0.05 < |z|$ and 0 is not in the interval $\therefore$ Reject H_0

Critical value at $\alpha = 0.01 > |z|$ and 0 is in the interval $\therefore$ fail to reject H_0

We reject the $H_0 : S = 0$ exactly when $|z|$ exceeds the critical value which is the same as 0 falling outside the corresponding confidence interval.

3) $$SE(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{3}{\sqrt{36}} = 0.5$$

95% confidence interval

$$\bar{x} \pm 1.96 \cdot SE = 11.5 \pm 1.96(0.5)$$

$= 11.5 \pm 0.98 = (10.52, 12.48)$

$$Z = \frac{\bar{x} - \mu_0}{SE} = \frac{11.5 - 12}{0.5} = -1.00$$

$|Z| = 1.00 < 1.96$ so we fail to reject $H_0: \mu = 12$ at $\alpha = 0.05$.

0 lies outside CI as well.

Yes, 12 is possible since the sample mean is only one standard error below the benchmark which is well within ordinary sampling variation.

Failing to reject does not prove that $\mu = 12$. The interval shows that values from 10.5 to 12.5 are all consistent with the data.

4) $\bar{X}_A - \bar{X}_B = 25 - 22 = 3$

$$SE(\bar{X}_A - \bar{X}_B) = \sqrt{\frac{S_A^2}{n_A} + \frac{S_B^2}{n_B}} = \sqrt{\frac{69}{100} + \frac{49}{100}}$$

$$= \sqrt{1.13} = 1.063$$

$$Z = \frac{(\bar{X}_A - \bar{X}_B) - 0}{SE} = \frac{3}{1.063} \approx 2.82$$

95% confidence Interval for $\mu_A - \mu_B$

$$3 \pm 1.96(1.063) = 3 \pm 2.08 = (0.92, 5.08)$$

$|Z| = 2.86$ exceeds 1.96, so we reject H_0 at $\alpha = 0.05$. The interval also agrees since 0 is not in the range.

5) $\bar{X}_A - \bar{X}_B = 14 - 12 = 2$

$$SE(\bar{X}_A - \bar{X}_B) = \sqrt{\frac{S_A^2}{n_A} + \frac{S_B^2}{n_B}} = \sqrt{\frac{36}{81} + \frac{25}{64}}$$

$$\approx 0.914$$

$$z = \frac{(\bar{X}_A - \bar{X}_B) - 0}{SE} = \frac{2}{0.914} \approx 2.19$$

6) $\bar{X}_A - \bar{X}_B = 3.2 - 2.8 = 0.4$

$$SE = \sqrt{\frac{S_A^2}{n_A} + \frac{S_B^2}{n_B}} = \sqrt{\frac{2.25}{49} + \frac{1.96}{64}}$$

$$\approx 0.277$$

$$z = \frac{0.4 - 0}{0.277} \approx 1.45$$

95% confidence interval

$0.4 \pm 1.96\,(0.277) = 0.4 \pm 0.54$

$= (-0.19, 0.99)$

at $\alpha = 0.05$: $|z| = 1.45$ is less than 1.96 so we fail to reject H_0.

The 95% CI interval has 0 in range.

No, the difference is not large relative to the noise ∴ the observed gap of 0.4 is only about 1.45 standard errors which is well within what sampling variation alone could produce.

7a) Has central air: 2734
Does not have central air : 196

Estimated portion with central air : 0.9331

b) Seed: 5030, number of bootstrap replications: 2000,

c) Bootstrap Standard error: 0.0046

d) 90% CI = [0.9255, 0.9407]
95% CI = [0.9241, 0.9422]
99% CI = [0.9212, 0.9450]

e) $\alpha = 0.10$, 90% CI [0.9255, 0.9407]: Reject H_0
$\alpha = 0.05$, 95% CI [0.9241, 0.9422]: Reject H_0
$\alpha = 0.01$, 99% CI [0.9212, 0.9450]: Reject H_0
∴ I reject H_0 at all significance levels ∵ 90% falls below the lower bounds of all 3 CI. This indicates that the data are not consistant with 90% of homes having central air, rather that true portion of homes with central air exceeds 90%

8) a) seed : 5030

Number of bootstrap replication: 2000

Mean of bootstrap estimates: 0.7229

b) Bootstrap standard error: 0.010092

c) 90% CI: [0.7066, 0.7398]

95% CI: [0.7034, 0.7429]

99% CI: [0.6972, 0.7492]

d) $\alpha = 0.10$, 90% CI [0.7066, 0.7398]: Reject H_0

$\alpha = 0.05$, 95% CI [0.7034, 0.7429]: Reject H_0

$\alpha = 0.01$, 99% CI [0.6972, 0.7492]: Reject H_0

I reject H_0 at all 3 α : 0 falls outsind the 90%, 95% and 99% CI.

This indicate a positive association between np.arcsinh(area) and np.arcsinh(price) where homes with larger transformed

area tend to have a higher tranformed price.

e) Area and price must remain paired during resampling because the correlation measures how the 2 variable move together within the same home. If area and price were sampled seperately, each bootstrap sample would match the area of one home with the price of another home, which would break the association in the data.

9a) seed: 5030

number of replications: 2000

b) bootstrap standard error : 2.1309 months

c) 90% CI: [125.5898, 132.5988] months

95% CI: [124.9187, 133.2699] months

99% CI: [123.6069, 134.5922] months

d) $\alpha = 0.10$, 90% CI [125.5898, 135.5988] : Reject H_0

$\alpha = 0.05$, 95% CI [124.9187, 133.2699]: Reject H_0

$\alpha = 0.01$, 99% CI [123.6069, 134.5922]: Reject H_0

e) The estimated mean recorded time to death or last follow up is 129.093 months $\approx$ 10.7579 years.

90% CI: [10.4658, 11.0499] years

95% CI: [10.4099, 11.1058] years

99% CI: [10.3005, 11.2152] years

These results do not estimate unrestricted mean life time because for patients

who were still living at their follow up overall survival (months) records only the time until that follow up rather than their eventual time of death, which means their true survival times are right censored and are at least as long as the recorded values. ∴ the mean of the recorded times treats these censored follow up times as if they were death times so that would understate the mean lifetime that would be observed if every patient were followed until death.

10) a) seed = 5030

number of replications = 2000

b) Bootstrap Standard error : 0.0233 hours

c) 90% CI : [7.5829, 7.6595] hours

95% CI : [7.5756, 7.6689] hours

99% CI : [7.5613, 7.6812] hours

d) $\alpha = 0.1$, 90% CI : Reject H_0

$\alpha = 0.05$, 95% CI : Reject H_0

$\alpha = 0.01$, 99% CI : Reject H_0

e) Estimate minus 8 hours : -0.3788 hours
= -22.7 minutes

90% CI for the gap : [-25.0, -20.4] min

95% " " " " : [-25.5, -20.0] "

99% " " " " : [-26.3, -19.1] "

statistically the result is clear since the bootstrap standard error

of 0.0233 hours is small relative to the gap, which leads interval to exclude 0 hours and H0 at all α. Practically however the gap ranges from 19.1 to 26.3 minutes per night across 90%, 95%, 99% intervals. So wether the 2 stories match depends on wether a difference of this size is meaningful for sleep